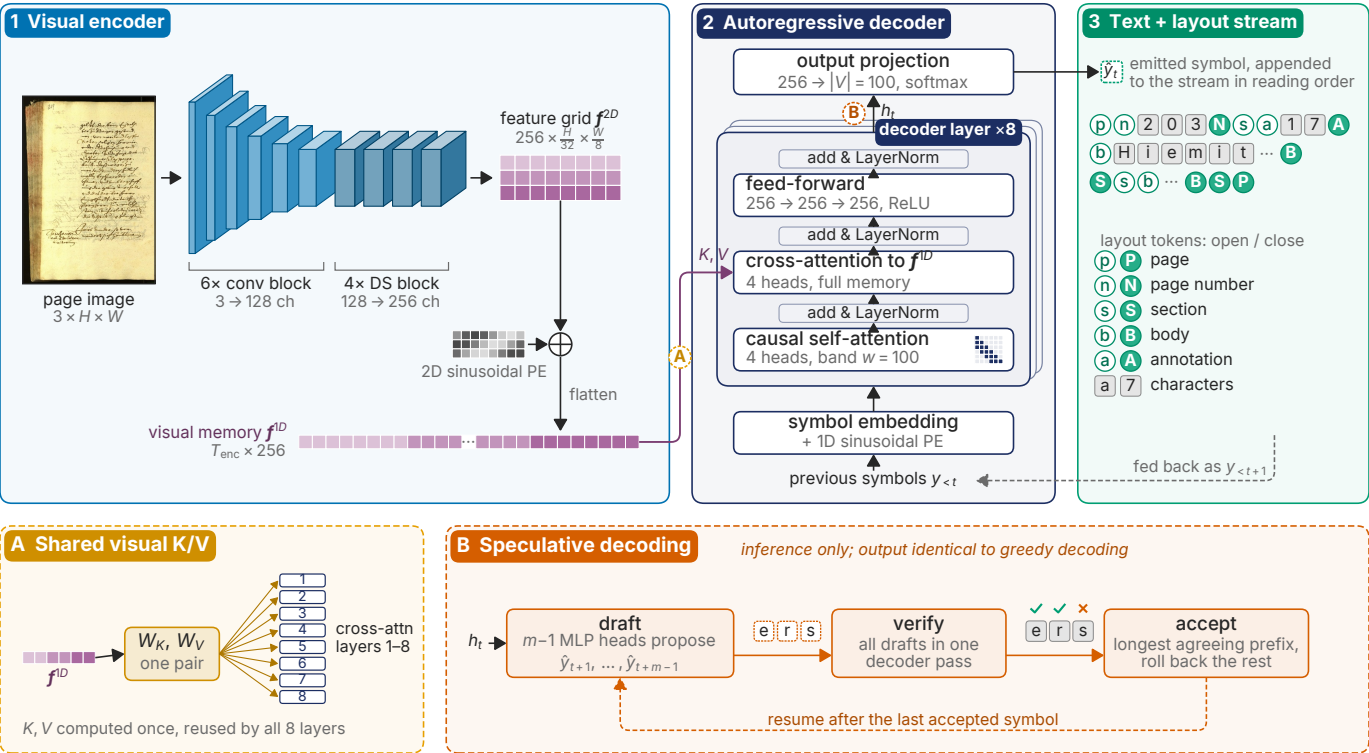

1 Visual encoder
page image
3 × H × W
6× conv block
3 → 128 ch
4× DS block
128 → 256 ch
feature grid f2D
256 × H/32 × W/8
2D sinusoidal PE
flatten
visual memory f1D
T_enc × 256
A
2 Autoregressive decoder
output projection
256 → |V| = 100, softmax
B
h_t
decoder layer ×8
add & LayerNorm
feed-forward
256 → 256 → 256, ReLU
add & LayerNorm
K, V
cross-attention to f1D
4 heads, full memory
add & LayerNorm
causal self-attention
4 heads, band w = 100
symbol embedding
+ 1D sinusoidal PE
previous symbols y_<t
3 Text + layout stream
ŷ_t
emitted symbol, appended to the stream in reading order
layout tokens: open / close
page
page number
section
body
annotation
characters
fed back as y_<t+1
A Shared visual K/V
f1D
W_K, W_V
one pair
cross-attn layers 1–8
K, V computed once, reused by all 8 layers
B Speculative decoding
inference only; output identical to greedy decoding
h_t
draft
m−1 MLP heads propose
ŷ_t+1, … , ŷ_t+m−1
verify
all drafts in one decoder pass
accept
longest agreeing prefix, roll back the rest
resume after the last accepted symbol